Chapter 14, Problem 56P

Problem

(a) Show that for a band-pass filter,

$$\mathbf{H}(s) = \frac{sB}{s^2 + sB + \omega_0^2}, \qquad s = j\omega$$

where B = bandwidth of the filter and $\omega 0$ is the center frequency.

(b) Similarly, show that for a band-stop filter,

$$\mathbf{H}(s) = \frac{s^2 + \omega_0^2}{s^2 + sB + \omega_0^2}, \qquad s = j\omega$$

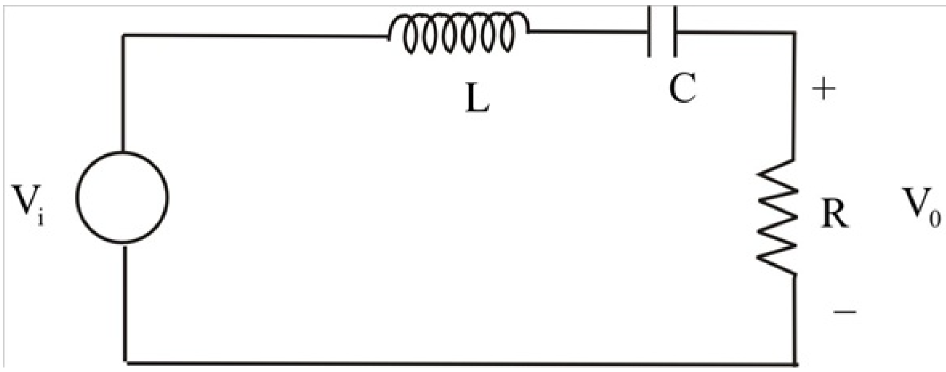
Step-by-step solution

Step 1 of 6

Consider Band pass filter

Step 2 of 6

(a)



Step 3 of 6

Transfer function is $H(\omega)=\frac{V_o}{V_1} \qquad -(1)$

In given circuit, transfer function is

$$H(\omega)=\frac{R}{R+j\left(\omega L-\frac{1}{\omega C}\right)} \qquad -(2)$$

$$or,\ H(\omega)=\frac{Rj\omega C}{Rj\omega C+j^2\omega^2 L+1} \qquad -(3)$$

Put Jω=.S. in (3) we get

$$H(\omega)=\frac{SRC}{SRC+S^2L^2+1}.$$

Divide by LC

$$H(\omega)=\frac{S\left(\frac{R}{L}\right)}{S\left(\frac{R}{L}\right)+S^2+\frac{1}{LC}} \qquad -(4)$$

$$we\ know,\ B=\frac{R}{L}, \quad and \quad \omega_o^{\ 2}=\frac{1}{LC} \qquad -(5)$$

From (4) *and* (5) *we get*

$$\boxed{H(\omega)=\frac{SB}{S^2+SB+\omega_o^{\ 2}}.}$$

Step 4 of 6

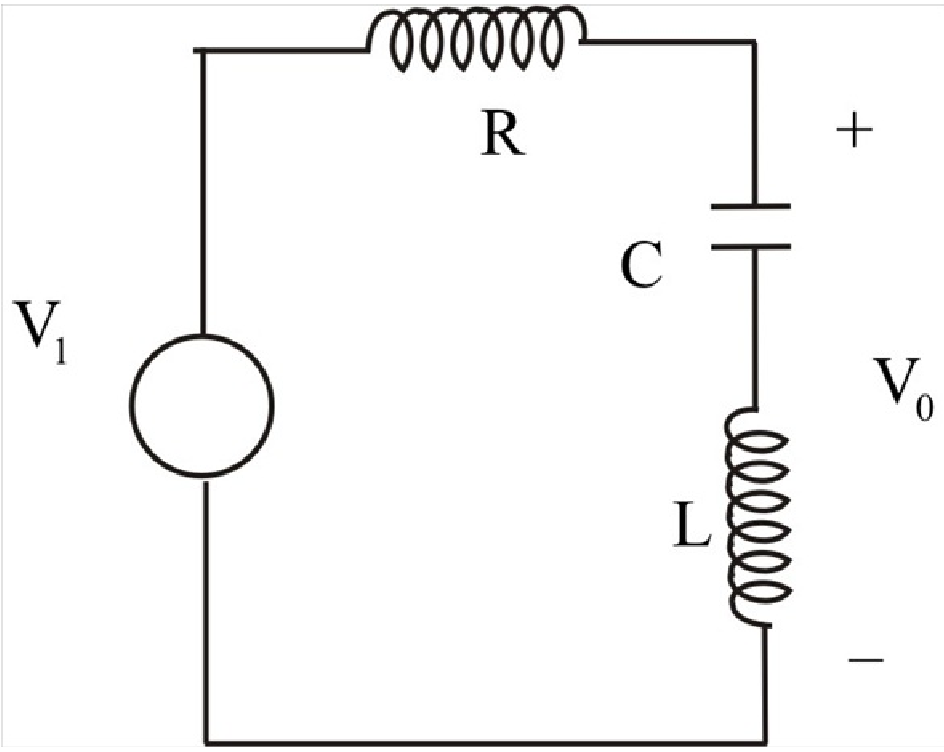
(b) Consider Band stop filter.

Transfer function is given by $\mathrm{H}(\omega)$

$$\mathrm{H}(\omega)=\frac{V_o}{V_i}=\frac{\left(j\omega L+\frac{1}{j\omega C}\right)}{R+j\omega L+\frac{1}{j\omega C}}$$

$$=\frac{j^2\omega^2 LC+1}{j\omega CR+j^2\omega^2 LC+1}$$

Step 5 of 6



Step 6 of 6

$$= \frac{\cancel{LC}\left(j^2\omega^2 + \frac{1}{LC}\right)}{\cancel{LC}\left(\frac{j\omega \cancel{C} R}{L\cancel{C}} + j^2\omega^2 + \frac{1}{LC}\right)}.$$

$$\boxed{H(\omega) = \frac{S^2 + \omega_o^{\ 2}}{SB + S^2 + \omega_o^{\ 2}}}$$ proved.

where, $S = j\omega \qquad B = \frac{R}{L} \qquad \omega_o^{\ 2} = \frac{1}{LC}$